

Article

Towards Machine Learning-Based FPGA Backend Flow: Challenges and Opportunities

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Abstract: Field-Programmable Gate Array (FPGA) is at the core of System on Chip (SoC) design across various Industry 5.0 digital systems—healthcare devices, farming equipment, autonomous vehicles and aerospace gear to name a few. Given that pre-silicon verification using Computer Aided Design (CAD) accounts for about 70% of the time and money spent on the design of modern digital systems, this paper summarizes the machine learning (ML)-oriented efforts in different FPGA CAD design steps. With the recent breakthrough of machine learning, FPGA CAD tasks—high-level synthesis (HLS), logic synthesis, placement and routing—are seeing a renewed interest in their respective decision-making steps. We focus on machine learning-based CAD tasks to suggest some pertinent research areas requiring more focus in CAD design. The development of open-source benchmarks optimized for an end-to-end machine learning experience, intra-FPGA optimization, domain-specific accelerators, lack of explainability and federated learning are the issues reviewed to identify important research spots requiring significant focus. The potential of the new cloud-based architectures to understand the application of the right ML algorithms in FPGA CAD decision-making steps is discussed, together with visualizing the scenario of incorporating more intelligence in the cloud platform, with the help of relatively newer technologies such as CAD as Adaptive OpenPlatform Service (CAOS). Altogether, this research explores several research opportunities linked with modern FPGA CAD flow design, which will serve as a single point of reference for modern FPGA CAD flow design.

Keywords: FPGA backend flow; machine learning; CAD design steps; synthesis; placement; routing



Citation: Taj, I.; Farooq, U. Towards Machine Learning-Based FPGA Backend Flow: Challenges and Opportunities. *Electronics* **2023**, *12*, 935. <https://doi.org/10.3390/electronics12040935>

Academic Editor: Wojciech M. Zabołotny

Received: 12 January 2023

Revised: 3 February 2023

Accepted: 7 February 2023

Published: 13 February 2023



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1. Introduction

Industrial Revolution 5.0 [1] holds the promise to elevate the overall quality of life, by using technology tools that create the infrastructure between systems and technologies. Today's sophisticated smart phones, autonomous vehicles, drones, FinTech applications, farming equipment and many other consumer applications are interconnected and involve massive data collection and processing using smart sensors. Each of these applications contain a System on Chip (SoC) with enormous computational capabilities. This enormous computational capability can be attributed to the improved design techniques that incorporate artificial intelligence, deep learning, augmented reality and digital twins. Field-Programmable Gate Arrays (FPGAs)—used for SoC prototyping—are at the core of this technology-driven revolution as they play a key role in sensor fusion and incoming data stream merging, thereby enhancing the overall productivity and efficiency. Figure 1 demonstrates the market growth rate of such consumer devices over the past few years [2]. However, FPGA designers face the challenge of long Computer-Aided Design (CAD) flow time required to map an application on the target FPGA. There are two reasons for this:

1. The increasing capacity of FPGA that could contain 10 million logic cells and 40 billion transistors [3];